

Git & GitHub Documentation

A Comprehensive Command Reference & Cheat Sheet

1. Git Setup & Configuration

Essential commands for first-time setup and identity configuration.

Command	Description	Example
<code>git --version</code>	Check installed Git version	<code>git --version</code>
<code>git config --global user.name "Name"</code>	Set global username	<code>git config --global user.name "Manu"</code>
<code>git config --global user.email "mail@example.com"</code>	Set global email	<code>git config --global user.email "abc@gmail.com"</code>
<code>git config --list</code>	Show all Git configurations	<code>git config --list</code>
<code>git help <command></code>	Get help for a command	<code>git help commit</code>

2. Repository Initialization

Command	Description	Example
<code>git init</code>	Initialize a new Git repository	<code>git init</code>
<code>git clone <url></code>	Clone a remote repository	<code>git clone https://github.com/user/repo.git</code>

3. File Tracking & Staging

Command	Description	Example
<code>git status</code>	Show repository status	<code>git status</code>
<code>git add file.txt</code>	Add specific file to staging area	<code>git add app.js</code>
<code>git add .</code>	Add all modified files	<code>git add .</code>
<code>git restore --staged file.txt</code>	Unstage file	<code>git restore --staged app.js</code>

Command	Description	Example
<code>git rm file.txt</code>	Delete tracked file	<code>git rm test.txt</code>
<code>git mv old new</code>	Rename/move file	<code>git mv old.txt new.txt</code>

4. Commit Operations

Command	Description	Example
<code>git commit -m "message"</code>	Create commit with message	<code>git commit -m "Added login page"</code>
<code>git commit -am "message"</code>	Add and commit tracked files	<code>git commit -am "Updated UI"</code>
<code>git log</code>	View commit history	<code>git log</code>
<code>git log --oneline</code>	Compact commit history	<code>git log --oneline</code>
<code>git show <commit-id></code>	Show commit details	<code>git show a1b2c3</code>

5. Branching Commands

Command	Description	Example
<code>git branch</code>	List branches	<code>git branch</code>
<code>git branch new-branch</code>	Create branch	<code>git branch feature-login</code>
<code>git checkout branch-name</code>	Switch branch	<code>git checkout dev</code>
<code>git checkout -b branch-name</code>	Create + switch branch	<code>git checkout -b feature-auth</code>
<code>git switch branch-name</code>	Modern branch switching	<code>git switch main</code>
<code>git merge branch-name</code>	Merge branch into current branch	<code>git merge dev</code>
<code>git branch -d branch-name</code>	Delete branch	<code>git branch -d feature-auth</code>

6. Remote Repository Commands

Command	Description	Example
<code>git remote -v</code>	View remotes	<code>git remote -v</code>
<code>git remote add origin <url></code>	Add remote repository	<code>git remote add origin https://github.com/user/repo.git</code>
<code>git remote add upstream <url></code>	Add upstream repo	<code>git remote add upstream https://github.com/original/repo.git</code>
<code>git remote rename old new</code>	Rename remote	<code>git remote rename origin upstream</code>
<code>git fetch</code>	Download remote changes	<code>git fetch origin</code>
<code>git pull origin main</code>	Fetch + merge changes	<code>git pull origin main</code>
<code>git push origin main</code>	Push changes to remote	<code>git push origin main</code>

7. Difference & Inspection Commands

Command	Description	Example
<code>git diff</code>	Show unstaged changes	<code>git diff</code>
<code>git diff --staged</code>	Show staged changes	<code>git diff --staged</code>
<code>git blame file.txt</code>	Show author history per line	<code>git blame app.js</code>
<code>git reflog</code>	Show reference logs	<code>git reflog</code>

8. Undo & Recovery Commands

Command	Description	Example
<code>git reset HEAD~1</code>	Remove latest commit	<code>git reset HEAD~1</code>
<code>git reset --hard HEAD~1</code>	Remove commit and changes	<code>git reset --hard HEAD~1</code>
<code>git revert <commit-id></code>	Create reverse commit	<code>git revert a1b2c3</code>
<code>git restore file.txt</code>	Restore file changes	<code>git restore app.js</code>

Command	Description	Example
<code>git stash</code>	Temporarily save changes	<code>git stash</code>
<code>git stash pop</code>	Restore stashed changes	<code>git stash pop</code>

9. Collaboration Workflow

Command	Description	Example
<code>git fetch upstream</code>	Fetch upstream changes	<code>git fetch upstream</code>
<code>git merge upstream/main</code>	Merge upstream main branch	<code>git merge upstream/main</code>
<code>git pull upstream main</code>	Pull upstream updates	<code>git pull upstream main</code>
<code>git request-pull origin/main mybranch</code>	Generate PR summary	<code>git request-pull origin/main mybranch</code>

10. GitHub CLI (gh) Commands

Install via WinGet: `winget install GitHub.cli`

Command	Description	Example
<code>gh auth login</code>	Login to GitHub	<code>gh auth login</code>
<code>gh repo create</code>	Create GitHub repository	<code>gh repo create myrepo</code>
<code>gh repo clone</code>	Clone repository	<code>gh repo clone user/repo</code>
<code>gh issue list</code>	View issues	<code>gh issue list</code>
<code>gh pr create</code>	Create pull request	<code>gh pr create</code>
<code>gh pr list</code>	List PRs	<code>gh pr list</code>
<code>gh pr merge</code>	Merge PR	<code>gh pr merge 1</code>
<code>gh workflow list</code>	View GitHub Actions workflows	<code>gh workflow list</code>

11. Advanced Git Commands

Command	Description	Example
<code>git cherry-pick <commit></code>	Apply commit from another branch	<code>git cherry-pick a1b2c3</code>
<code>git rebase main</code>	Reapply commits on another base	<code>git rebase main</code>
<code>git squash</code>	Combine commits during rebase	<code>git rebase -i HEAD~3</code>
<code>git tag v1.0</code>	Create version tag	<code>git tag v1.0</code>
<code>git clean -fd</code>	Remove untracked files	<code>git clean -fd</code>
<code>git bisect</code>	Find problematic commit	<code>git bisect start</code>